

Aula 07/15

Escola Tecnológica FPFtech

Curso: Automação para Indústria 4.0

Módulo: IoT Aplicado à Manufatura

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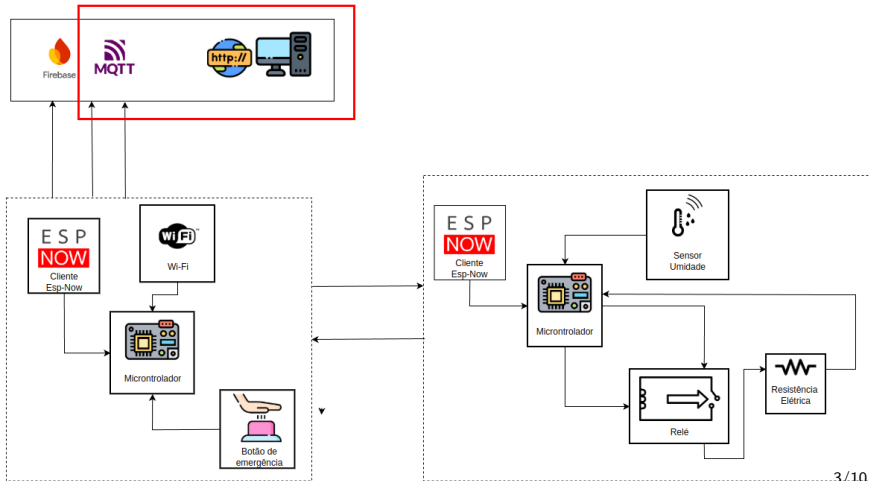
08 de abril de 2026

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Projeto Guiado

Fase 3:



Conectar no Wi-Fi

```
1 #include <WiFi.h>
2
3 const char* ssid = "SEU_WIFI";
4 const char* password = "SUA_SENHA";
5
6 void setup() {
7     Serial.begin(115200);
8
9     WiFi.mode(WIFI_STA);
10    WiFi.begin(ssid, password);
11
12    Serial.print("Conectando ao Wi-Fi");
13
14    while (WiFi.status() != WL_CONNECTED) {
15        delay(500);
16        Serial.print(".");
17    }
18
19    Serial.println("\nConectado!");
20    Serial.print("IP: ");
21    Serial.println(WiFi.localIP());
22 }
23
24 void loop() {
25 }
```

Envio HTTP com autenticação

```
1 #include <WiFi.h>
2 #include <HTTPClient.h>
3
4 const char* serverURL = "http://seu-servidor.com/api";
5
6 void setup() {
7     Serial.begin(115200);
8 }
```

Envio HTTP com autenticação

```
1 void loop() {
2   if (WiFi.status() == WL_CONNECTED) {
3     HTTPClient http;
4     http.begin(serverURL);
5     http.addHeader("Content-Type", "application/json");
6     http.setAuthorization("usuario", "senha");
7     String json = "{\"valor\": 42}";
8     int httpResponseCode = http.POST(json);
9     Serial.print("Código HTTP: ");
10    Serial.println(httpResponseCode);
11    if (httpResponseCode > 0) {
12      String response = http.getString();
13      Serial.println(response);
14    }
15    http.end();
16  }
17  delay(5000);
18 }
```

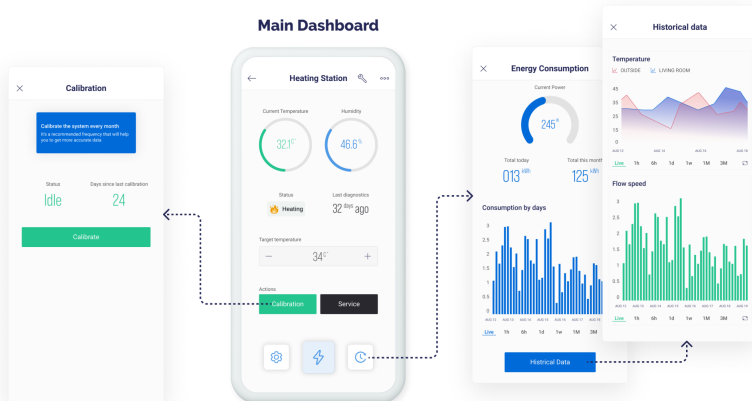
Envio MQTT

```
1 #include <WiFi.h>
2 #include <PubSubClient.h>
3
4 const char* mqtt_server = "broker.hivemq.com"; // broker p blico
5 const int mqtt_port = 1883;
6
7 WiFiClient espClient;
8 PubSubClient client(espClient);
9
10 void reconnect() {
11     while (!client.connected()) {
12         Serial.print("Conectando MQTT...");
13
14         if (client.connect("ESP32Client")) {
15             Serial.println("Conectado!");
16         } else {
17             Serial.print("Falhou, rc=");
18             Serial.print(client.state());
19             Serial.println(" tentando novamente...");
20             delay(2000);
21         }
22     }
23 }
```

Envio MQTT

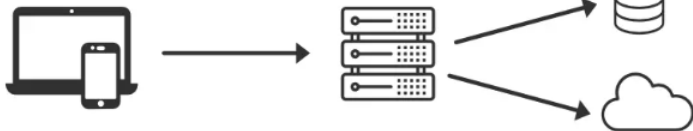
```
1 void setup() {  
2     Serial.begin(115200);  
3     client.setServer(mqtt_server, mqtt_port);  
4 }  
5  
6 void loop() {  
7     if (!client.connected()) {  
8         reconnect();  
9     }  
10  
11     client.loop();  
12  
13     String mensagem = "Hello MQTT!";  
14     client.publish("esp32/teste", mensagem.c_str());  
15  
16     Serial.println("Mensagem enviada");  
17  
18     delay(5000);  
19 }
```


Projeto Guiado



Projeto Guiado

Traditional



Firebase



Firebase

Quiz de Revisão

